The examination is being carried out on the **following application documents**

Description, Pages

1-34 filed with entry into the regional phase before the EPO

Claims, Numbers

1-7 filed with entry into the regional phase before the EPO

Drawings, Sheets

1/5-5/5 filed with entry into the regional phase before the EPO

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 US 5 564 152 A (JEANNET ROLAND [DE] ET AL) 15 October 1996 (1996-10-15)

D3 WO 2005/039355 A1 (LION CORP [JP]; KATO TAKAO [JP] ET AL.) 6 May 2005 (2005-05-06)

The following document has been cited in the international search report; the numbering will be adhered to in the rest of the procedure.

D2 WO 00/49911 A1 (INOUE, Susumu) 31 August 2000 (2000-08-31)

1 Novelty and Inventive Step (Art. 52(1), 54 and 56 EPC)

1.1 Independent claim 1

The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claim 1 does not involve an inventive step within the meaning of Article 56 EPC.

1.1.1 Document D1 is considered to be the prior art closest to the subject-matter of claim 1 and discloses (references between brackets apply to this document):

A toothbrush (*the toothbrush of fig.1, reference sign 20*) comprising:

a handle body made of a hard resin (*fig.1, reference sign 24 shows and handle. It is implicit, for the person skilled in the art, that the handle is made of hard resin. Otherwise, the operability of the toothbrush will be compromised*);

and
a brush moulded body ~~made of a soft resin~~ and provided on a distal end side of the handle body (*fig.1, reference sign 22, shows the brush head. Column 4, lines 61-67 and col.5 lines 1-6 disclose that the brush head can be moulded*),

wherein the handle body has an insertion portion provided to protrude to the distal end side (*fig.3 and col.3, lines 11-14 show a "threaded pin 26...which projects from the front end of the brush handle 24"*),

the brush moulded body has a head base portion (*fig.3, reference sign 33, "bristle bed"*) and a filament (*fig.3, reference sign 29, "bristles"*) configured to protrude to a front surface side of the head base portion,

the head base portion has an insertion hole (*fig.3, reference sign 28, "integral treated hole" moulded integrally - col.5 lines 1-6 - with the brush head)* into which the insertion portion is inserted and fitted,

the brush moulded body has a maximum distance between a base end of the filament and a back surface of the head base portion of ~~2.0 mm or more and 4.5 mm or less~~, and

when a maximum length of the head base portion in an insertion direction is defined as L1 and a maximum length of the insertion hole in the insertion direction is defined as L2, a ratio of the maximum length of the insertion hole to the maximum length of the head base portion, which is represented by L2/L1, ~~is 0.30 or more and 0.70 or less~~.

1.1.1.1 The toothbrush of D1 does not disclose that:

i) the brush moulded body is made of soft resin

ii) the thickness of the head is between 2 and 4.5 millimetres

iii) the depth of the insertion hole is between 30 and 70% of the length of the brush moulded body.

1.1.1.2 The technical effects related to these features are:

i) preventing breakage of the brush moulded body. Said brush moulded body is deformed under the action of the insertion portion, made of hard material, when the latter is coupled with the insertion hole (par.58 of the present application)

ii) the limited thickness improves the operability of the brush moulded body when inserted in the mouth (par.44 of the present application)

iii) the brush moulded body is stiffened by the insertion portion thus, once coupled, it is not easily bendable (par.48 of the present application)

1.1.1.3 The objective technical problems are:

i) allow a secure and durable connection between the brush moulded body and the handle

ii) improve operability of the brush moulded body

iii) improve the stiffness of the brush moulded body

1.1.1.4 The person skilled in the art, trying to solve the objective technical problems, would consider the teaching of D1 for problems *i), iii)* and the teaching of D3 for problem *ii)*. D1 proposes an alternative solution to connect the brush moulded body and the handle: a threaded pin - fig.3, reference sign 26 - fixed on the handle is connectable to a threaded hole - fig.3, reference sign 28 - realized on the brush moulded body made of plastic (col.5, lines 3-10). It is worth noting that the brush moulded body is implicitly made of hard plastic so to allow the realization of a threaded connection. This type of connection does not require the brush moulded body to deform, thus avoiding the onset of dangerous stresses that could damage its structure. At the same time, the brush moulded body does not need to be made of a flexible material, prone to be bended, but is already provided with a suitable mechanical characteristics for brushing teeth.

On the other hand, it is well know in the art that a thick brush moulded body could negatively impact the operability of the toothbrush. For example, D3 discloses a moulded toothbrush (fig.6a-6c) preferably having the thickness of the head between 2-6 mm (par.56) so to increase operability.

Therefore the subject matter of claim 1 is not inventive in view of the disclosure of D1 and D3.

1.1.2 Alternatively, D2 can be considered as the closest prior art to the subject-matter of claim 1. Document D2 discloses (references between brackets apply to this document):

A toothbrush (*the toothbrush of fig.1-5*) comprising:

a handle body made of a hard resin (*fig.5(2) and par.28 show a handle made of plastic resin. It is implicit, for the person skilled in the art, that the handle is made of hard resin.*);

and
a brush moulded body made of a soft resin and provided on a distal end side of the handle body (*fig.5(3) and par.29 show a rubber head. Said head comprises pyramidal bristles. Such shape is implicitly obtainable with a moulding process.*),

wherein the handle body has an insertion portion provided to protrude to the distal end side (*fig.2 and par.35 disclose an insertion portion of the handle with a "saw tooth shape"*),

the brush moulded body has a head base portion and a filament configured to protrude to a front surface side of the head base portion (*fig.5(1) shows the head of the toothbrush with bristles*),

the head base portion has an insertion hole into which the insertion portion is inserted and fitted (*fig.5(1)-5(2) show an insertion hole on the head of the toothbrush. The coupling of said hole with the insertion portion of the handle forms a "junction...that can be firmly bonded without using the adhesive at the joint" - par.35)*,

the brush moulded body has a maximum distance between a base end of the filament and a back surface of the head base portion of ~~2.0 mm or more and 4.5 mm or less~~, and

when a maximum length of the head base portion in an insertion direction is defined as L1 and a maximum length of the insertion hole in the insertion direction is defined as L2, a ratio of the maximum length of the insertion hole to the maximum length of the head base portion, which is represented by L2/L1, ~~is 0.30 or more and 0.70 or less~~.

1.1.2.1 D2 does not disclose neither the thickness of the base nor the depth of the insertion hole.

1.1.2.2 The technical effects related to these features are:

i) the limited thickness improves the operability of the brush moulded body when inserted in the mouth (par.44 of the present application)

ii) the brush moulded body is stiffened by the insertion portion thus, once coupled, it is not easily bendable (par.48 of the present application).

1.1.2.3 The objective technical problems are:

i) improve operability of the brush moulded body

ii) improve the stiffness of the brush moulded body

1.1.2.4 The person skilled in the art, trying to solve the objective technical problems, would consider the teaching of D2 for problem *ii)* and the teaching of D3 for problem *i).* As already discusses under point 1.1.2, D2 discloses a toothbrush made in two parts: a handle made of plastic and a brush head made of rubber. The person skilled in the art would implicitly expect that the head of the toothbrush, after connection with the rigid handle, would be stiffened. Moreover, fig.5 shows that the length of the insertion portion is approximately the length of the head of the brush. Therefore, no major bending of the head is possible

On the other hand, it is well know in the art that a thick brush moulded body could negatively impact the operability of the toothbrush. For example, D3 discloses a moulded toothbrush (fig.6a-6c) preferably having the thickness of the head between 2-6 mm (par.56) so to increase operability.

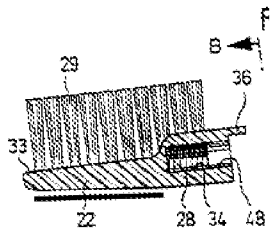
Therefore the subject matter of claim 1 is not inventive in view of the disclosure of D2 and D3.

1.2 Dependent claims 2-7

Dependent claims 2-7 do not appear to contain any additional features which, in combination with the features of any claim to which they refer, meet the requirements of the EPC with respect to inventive step.

1.2.1 Claim 2 defines a full portion of the brush moulded body. The technical effect of said feature corresponds with the technical effect discussed under point 1.1.1, i.e. the brush moulded is not easy bendable (par.83 of the present application). The objective technical problem is also the same as under point 1.1.1, i.e. improve the stiffness of the brush moulded body. The toothbrush from D1 discloses an alternative solution to solve the objective technical problem. Therefore, claim 2 cannot be considered as involving an inventive step (see point 1.1.1 above).

1.2.2 Claim 3 defines an inclined portion of the head base portion: see D1, fig.3 (reported below with a solid line highlighting the inclined portion of the head of the toothbrush)



1.2.3 Claims 4-5 define a type of connection between the toothbrush handle and the toothbrush head: see D1. fig.3 and point 1.1.1 where an alternative type of connection has been disclosed.

1.2.4 Claims 6 defines polyurethane as a possible material to be employed for the brush moulded body. This is a common material used for the manufacturing of toothbrushes: see for example D3, par.57.

1.2.5 Claim 7 defines a partition surface of the brush moulded body. This feature is merely one of several straightforward possibilities which the skilled person would select, depending on the circumstances, without exercising inventive skill, in order to mould a toothbrush head.

2 Other deficiencies

2.1 To meet the requirements of Rule 42(1)(b) EPC, D1 should be identified in the description and the relevant background art disclosed therein should be briefly discussed.

2.2 Independent claim 1 is not in the two-part form in accordance with Rule 43(1) EPC, which in the present case would be appropriate, with those features known in combination from D1 (see point 1 above) being placed in the preamble (Rule 43(1)(a) EPC) and the remaining features being included in the characterising part (Rule 43(1)(b) EPC).

Independent claim 1 should therefore be redrafted accordingly. If, however, the applicant is of the opinion that the two-part form would be inappropriate, reasons therefor should be provided. In addition, the applicant should ensure that it is clear from the description which features of the subject-matter of claim 1 are already known from D1 (see Guidelines F-IV, 2.3.2).

2.3 The features of the claims should be provided with reference signs placed in parentheses to increase the intelligibility of the claims (Rule 43(7) EPC). This applies to both the preamble and characterising portion (see Guidelines F-IV, 4.18).

3 Procedure

3.1 The applicant is invited to file new claims which take account of the above comments.

3.2 In order to comply with the requirements of Rule 137(4) EPC, the applicant should clearly identify the amendments made, irrespective of whether they concern amendments by addition, replacement or deletion, and indicate the passages of the application as filed on which these amendments are based (see Guidelines H III, 2.1).

3.3 Amendments should be made by filing replacement pages. Unnecessary recasting of the description should be avoided. An amended abstract is not required. The applicant should also take account of the requirements of Rule 50(1) EPC. According to Rule 50(1) in conjunction with Rule 49(8) EPC, amendments shall be typed or printed. The attention of the applicant is drawn to the fact that the application may not be amended in such a way that it contains subject-matter which extends beyond the content of the application as filed (Article 123(2) EPC). In order to assess the question of inventive step, the applicant is asked to indicate in the response which technical problem is solved by the characterising features of the new independent claim compared to the closest prior art.